

CIR 1차 세미나

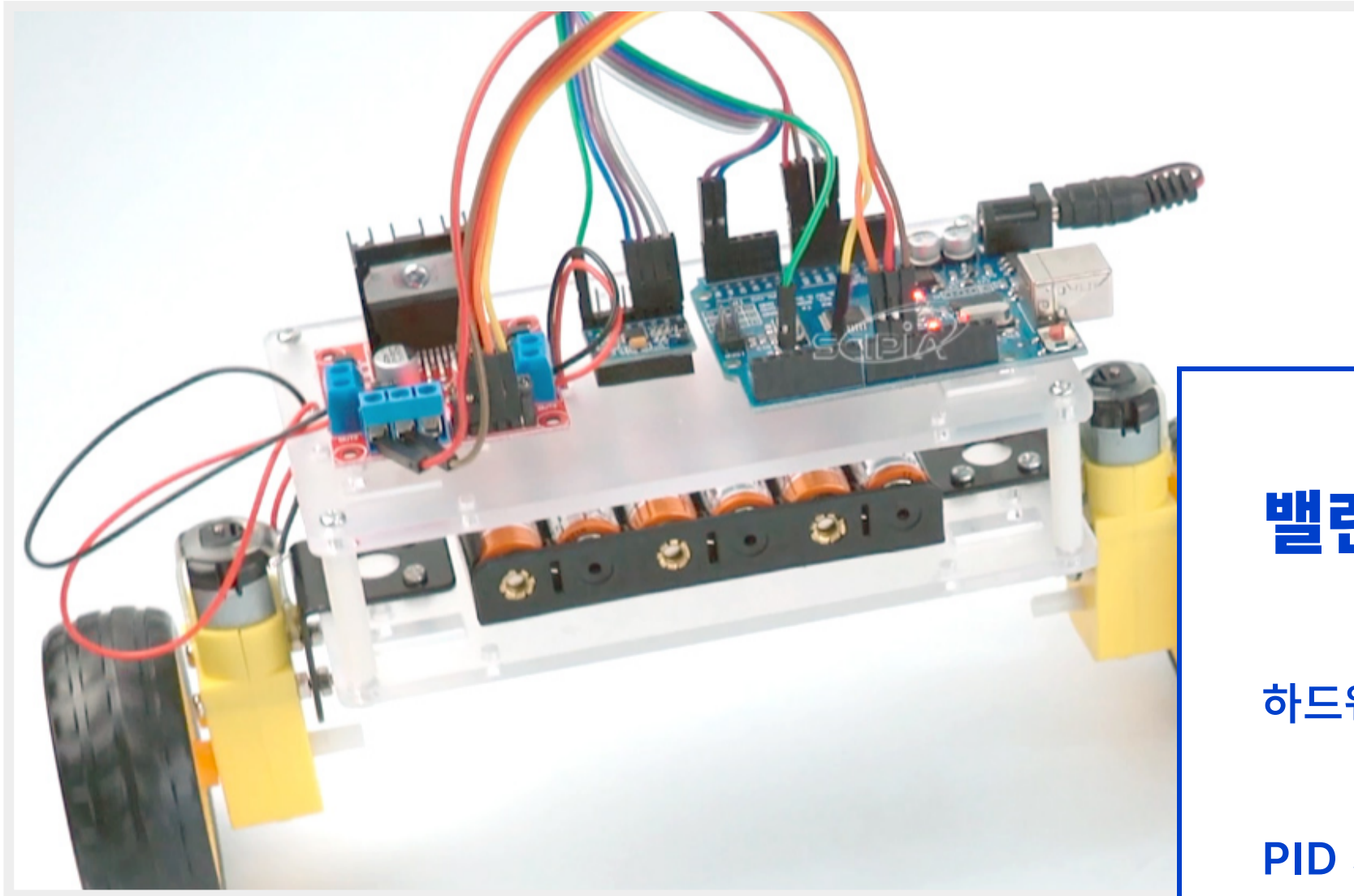
밸런싱 로봇

2025.01.09

메카트로닉스공학과 2021134006 김동진



1	_____	목적
2	_____	부품
3	_____	MPU 6050
4	_____	PID
5	_____	시스템 구성도
6	_____	flow chart
7	_____	계획



밸런싱 로봇

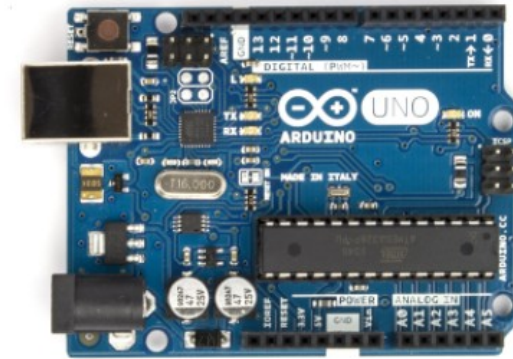
하드웨어의 설계 및 시스템 통합

PID 기반의 모터 제어

두 바퀴를 이용하여 넘어지지 않고 균형을 잡는 것을 목표로 함.

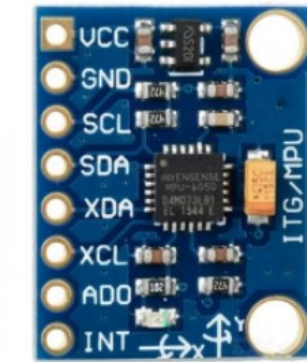
제어기

Arduino



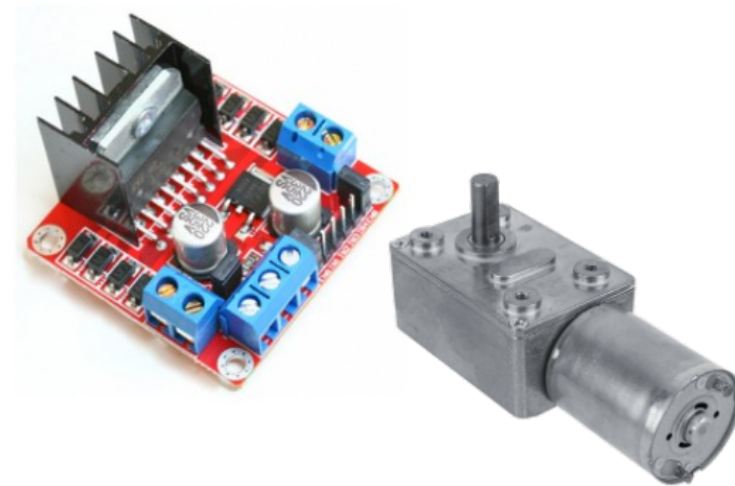
센서

MPU 6050



구동부

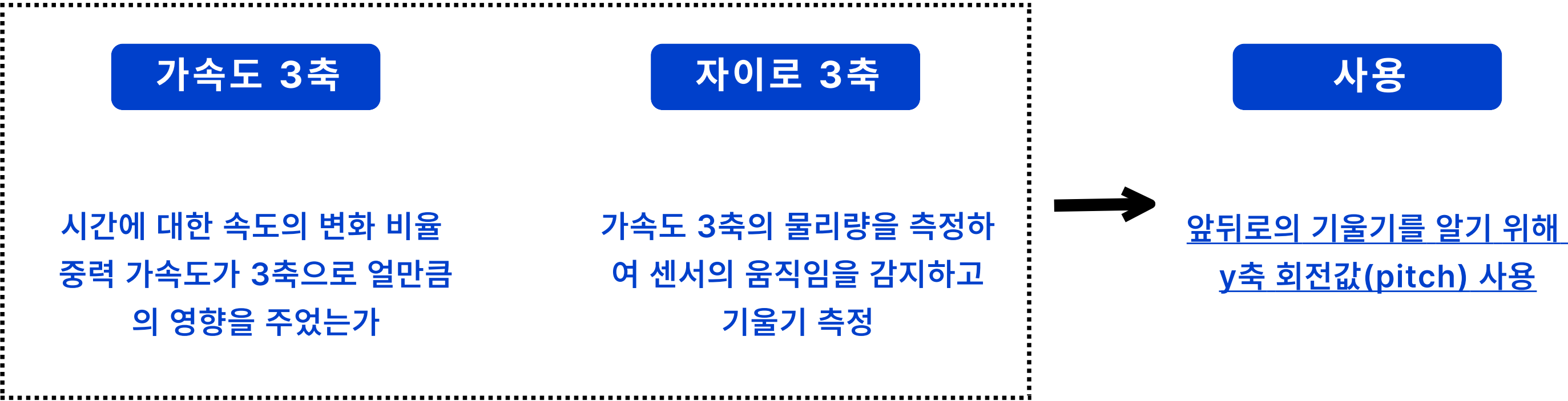
L298N 모터 드라이버
WGM40M-3157-1260 기어모터



전원

11.1V Li-Po 배터리







$$Output(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

Kp

비례동작

기울어진 각도만큼 돌아오는 힘

Ki

적분 동작

누적된 미세 오차를 제거

Kd

미분 동작

기울어진 속도에 저항하여 멈춤

5 시스템 구성도

전원

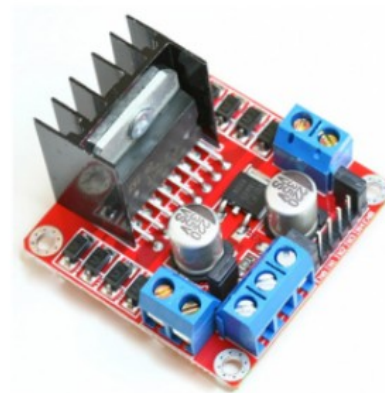
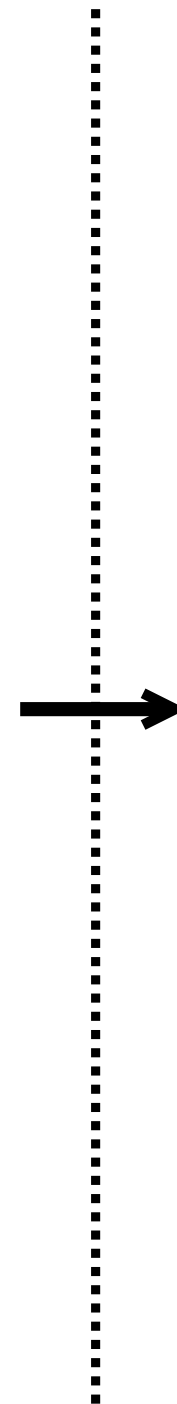
구동부

제어기

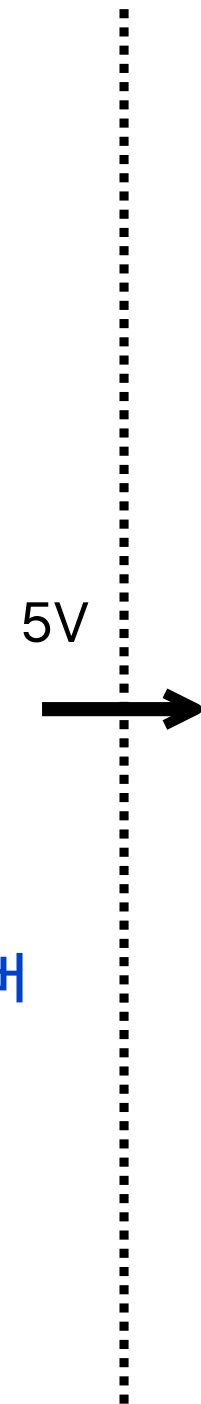
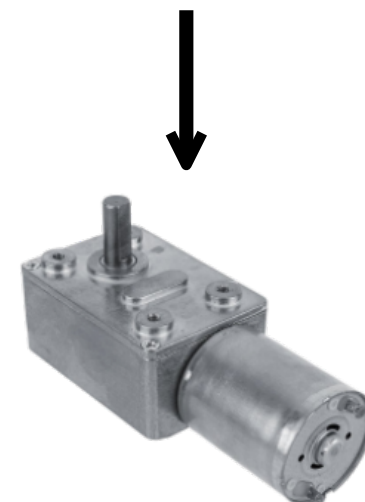
센서



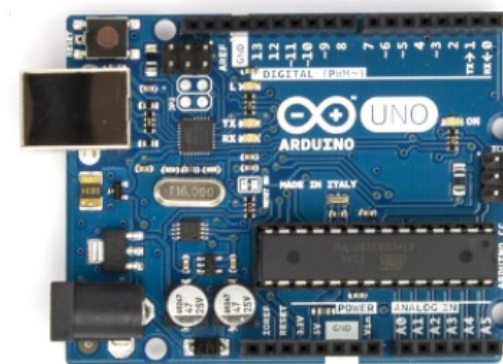
11.1V Li-Po 배터리



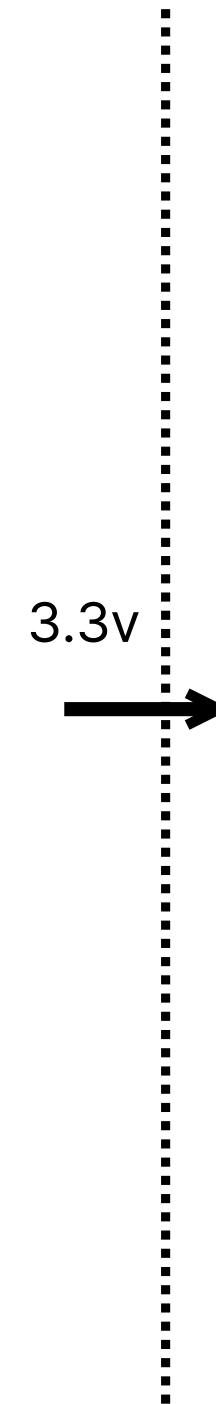
L298N 모터드라이버



5V



Arduino



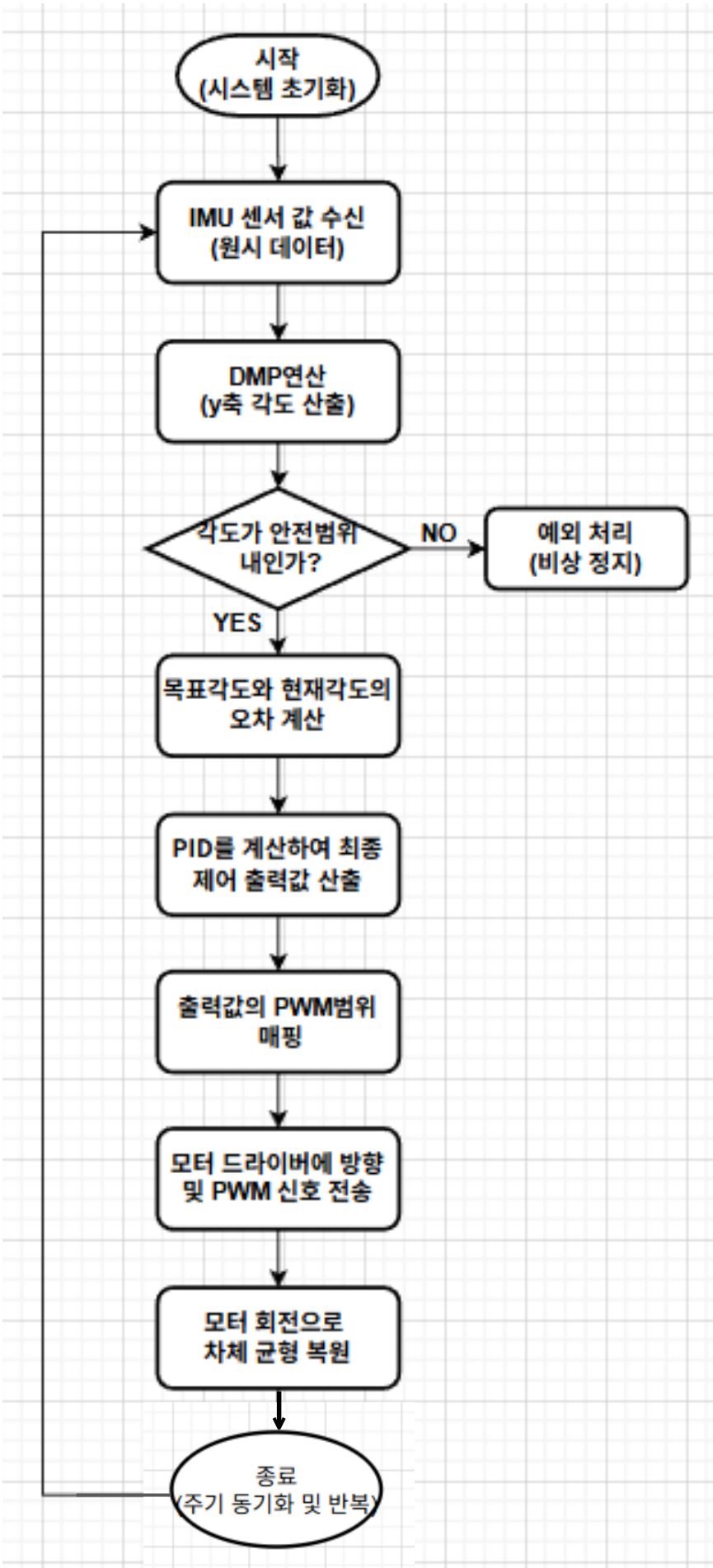
3.3v

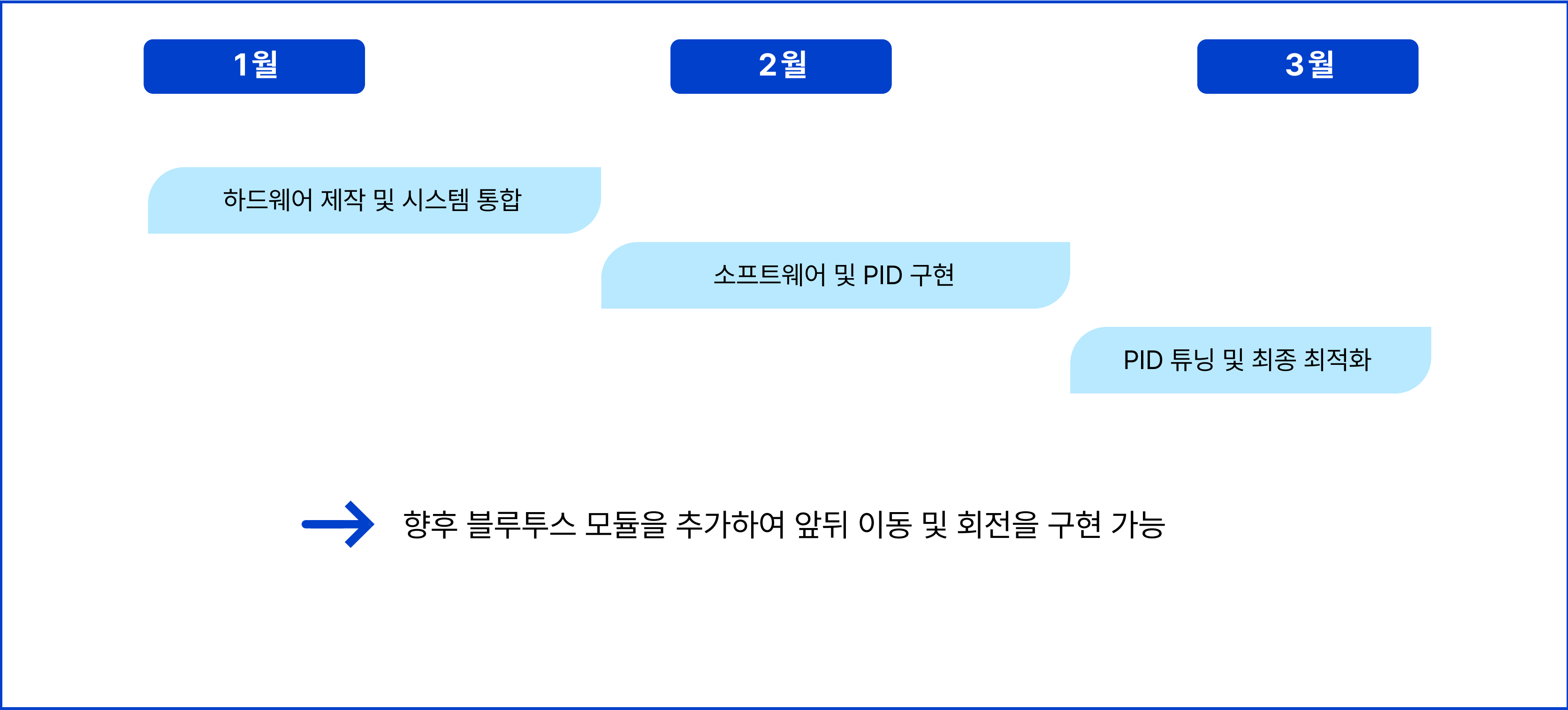


IMU 센서
(MPU 6050)



6 flow chart





감사합니다